

1. Conversion of Entity to DTO perform in service layer only
2. stream library - for conversion between collections
3. ModelMapper: Runtime reflection → Unsafe calls → Warnings in Java 21+
4. MapStruct: Compile-time code generation → Pure Java → No warnings, no overhead
5. Industry Standard: MapStruct > ModelMapper

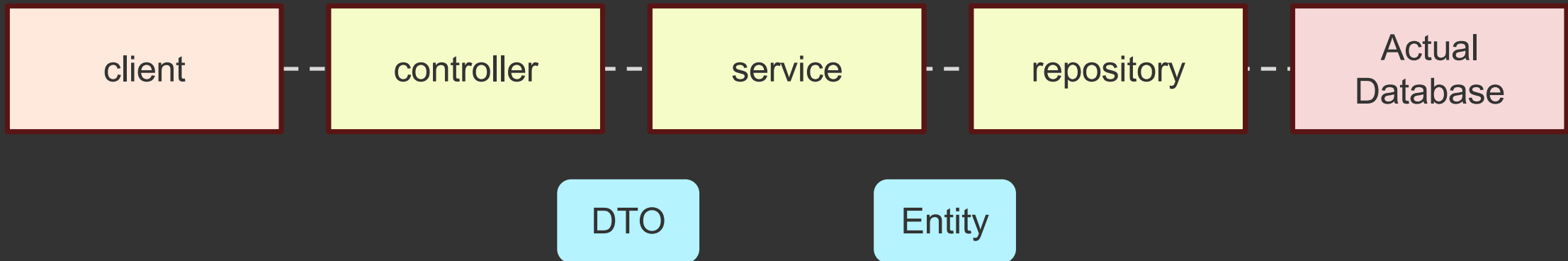
2.4

Service Layer



Using @Service create service classes inside Service Package - EmployeeService
Inject service inside controller using Constructor DI for usage
Perform task - findById() , findAll() , save() - inside controller
Create implementation for the declare method - returns DTO type not Entity
Add ModelMapper dependency from maven repo and create it's bean in config pkg - MapperConfig class
Use ModelMapper for conversion b/w Entity & DTO - modelmapper.map(entity, dto)
Using Stream Library to convert List<Entity> to List<DTO> - map() , lambda & collect(Collectors.toList())

Spring Boot Web Project Structure



Key roles of Service Layer

- The service layer acts as a bridge between the persistence layer (responsible for data access) and the presentation layer (handling user interaction)
- It encapsulates the business logic of the application, orchestrates interactions between different components, and provides a clean interface for external clients to interact with the system.
- By abstracting away the complexities of data access and business operations, the service layer promotes modularity, maintainability, and scalability.



2.5

PUT, PATCH and DELETE Mappings

